

ECEN 620

Homework #2

Due: 9-29-2026, 11:59PM

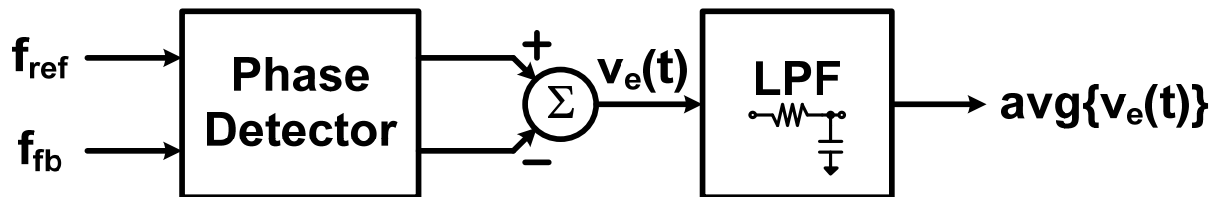
Homeworks will not be received after due.

Instructor: Sam Palermo

This homework requires transistor-level circuit design for Problem 2 & 3. You may use any CMOS technology to solve the problem, as long as it is a 90nm or more advanced technology node (shorter channel length). For students who do not have access to a design kit, instructions on how to access the default 90nm CMOS transistor models are posted on the website. For this 90nm technology assume a nominal 1.2V supply.

1. Phase Detector Characterization.

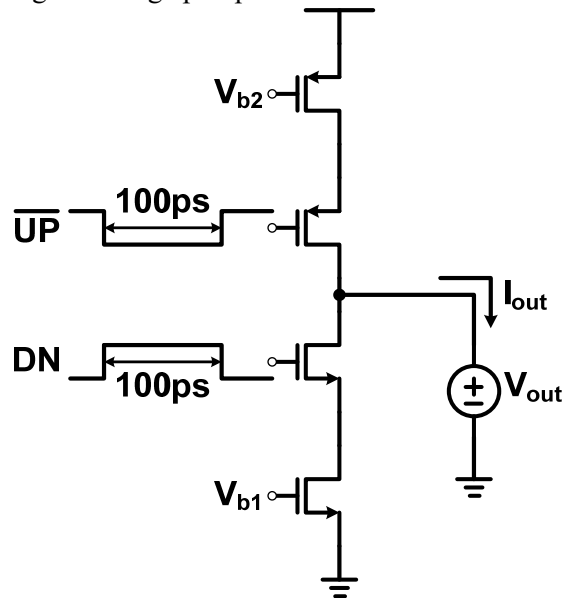
- a. For the 4 phase detectors discussed in Lecture 4 (Analog Mixer, XOR, JK Flip-Flop, PFD), produce the phase detector transfer curves ($\text{avg}(v_e)$ vs ϕ_e). Find the phase detector gain at the nominal PLL lock point.
 - i. Use the test setup shown in the figure below with $f_{fb}=f_{ref}$. For the analog mixer, use sinusoidal clock inputs. For all the others, use CMOS-level square-wave clocks. Use a VCVS as a buffer between the PD output and the averaging filter. For PDs with 2 outputs, a multi-input VCVS can be used to subtract the 2 outputs before low-pass filtering.
 - ii. Plot over a phase range of $\pm 4\pi$, with a minimum of 20 phase points per curve.
 - iii. Feel free to use a macro-model approach to generate the phase detector transfer curves, i.e. no transistor-level design is required. Although, if you prefer to implement the circuits at the transistor level, feel free to do so.
- b. For the 4 phase detectors, find the average output with $f_{fb}=0.5, 0.75, 1.5$, and $2f_{ref}$. Assume an initial phase difference of 0° . What do these values imply regarding the usefulness of these circuits as a frequency detector?



Phase Detector Characterization Test Bench

2. **Phase-Frequency Detector Design.** Design at the transistor level a PFD working at 100MHz with a minimum PFD output pulse width of at least 100ps.
 - a. Give the transistor-level schematic and simulations showing the minimum PFD pulse width specification is satisfied.
 - b. Plot the phase transfer characteristic over a phase range of $\pm 4\pi$, with a minimum of 20 phase points per curve.

3. **Charge Pump Design.** Design at the transistor level a simple charge pump (Lecture 5, Slide 8) that can supply a nominal $I_{CP}=100\mu A$ and operate with the PFD from Problem 2. **Use one ideal reference current source and appropriate current mirrors to produce the charge pump bias voltages (V_{b1} & V_{b2}).**
- Use the test setup shown in the figure below with ideal 100ps switch signals that are in-phase. Use an ideal voltage source at the charge pump output to sense the net output current, I_{out} . From the transient I_{out} response, plot the peak (absolute value) and the average I_{out} as V_{out} is swept from $0.2*V_{DD}$ to $0.8*V_{DD}$. Compute the phase error produced by the average I_{out} value.
 - Repeat part (a) using the PFD designed in Problem 2 to produce the charge pump switch signals. Assume the PFD inputs are perfectly in-phase.
 - Implement appropriate design techniques to improve both the peak and average I_{out} values, assuming the PFD is driving the charge pump.



Simple Charge Pump Characterization Test Bench